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I'm Something of a ...



Monet painting - G...



MSP Monet Notebo...



P8\_GAN\_032023



Photo to Monet — ...



RECENTLY EDITED



[TF Demo] Image c...



Image Captioning b...



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## I'm Something of a Painter Myself

Will you be the next Monet?

going

Discussion Leaderboard Rules Team

### MiFID

Submissions are evaluated on MiFID (Memorization-informed Fréchet Inception Distance), which is a modification from [Fréchet Inception Distance \(FID\)](#).

The smaller MiFID is, the better your generated images are.

### What is FID?

Originally published [here \(github\)](#), FID, along with [Inception Score \(IS\)](#), are both commonly used in recent publications as the standard for evaluation methods of GANs.

In FID, we use the Inception network to extract features from an intermediate layer. Then we model the data distribution for these features using a multivariate Gaussian distribution with mean  $\mu$  and covariance  $\Sigma$ . The FID between the real images  $r$  and generated images  $g$  is computed as:

$$FID = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

where  $\text{Tr}$  sums up all the diagonal elements. FID is calculated by computing the Fréchet distance between two Gaussians fitted to feature representations of the Inception network.

